

Surviving above the snowline

THE HIGHER THE ALTITUDE at which a plant grows, the colder the temperatures it has to endure. Very low temperatures create specific problems for plant life. Thin mountain air holds little heat, and on exposed mountainsides, strong winds create a chill factor that makes the cold even more penetrating. In addition, low rainfall and thin, frozen soils mean that water is scarce. However, many plants manage to survive despite the harsh conditions. In the Himalayas of Asia, flowering plants have been found at more than 20,000 ft (6,000 m), sheltering in hollows in the frost-shattered rock. These plants, called alpine plants, are typically small and compact, so they can survive on the high mountain peaks, or in the frozen polar regions. Alpine plants often grow in dense cushions or flattened mats, giving protection against the cold, drying wind. Upright, spreading branches would be battered by the wind, and large leaves would lose valuable heat and water.



QUICK WORK

When spring comes, the mountain slopes burst into color as alpine plants begin to flower. In high mountain areas where the summers are short, these plants have to flower and produce seeds quickly before winter comes around again.

RADIATION HAZARD

The sunlight that falls on high mountaintops in the tropics is more intense than anywhere else on Earth. The silversword grows in Hawaii at altitudes of up to 15,000 ft (4,000 m). Its leaves are covered with fine white hairs that protect the plant from much of the Sun's dangerous ultraviolet radiation.

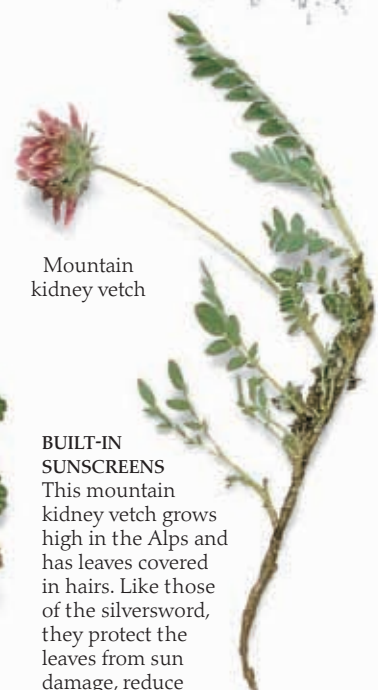


Mountain avens



HAIRY LEAVES
Mountain avens are plants that grow on high ground, from the Alps of Europe to the Arctic. Fine hairs on the undersides of their leaves minimize water loss and protect against the cold.

Mountain kidney vetch



BUILT-IN SUNSCREENS

This mountain kidney vetch grows high in the Alps and has leaves covered in hairs. Like those of the silversword, they protect the leaves from sun damage, reduce water loss, and act as insulation.

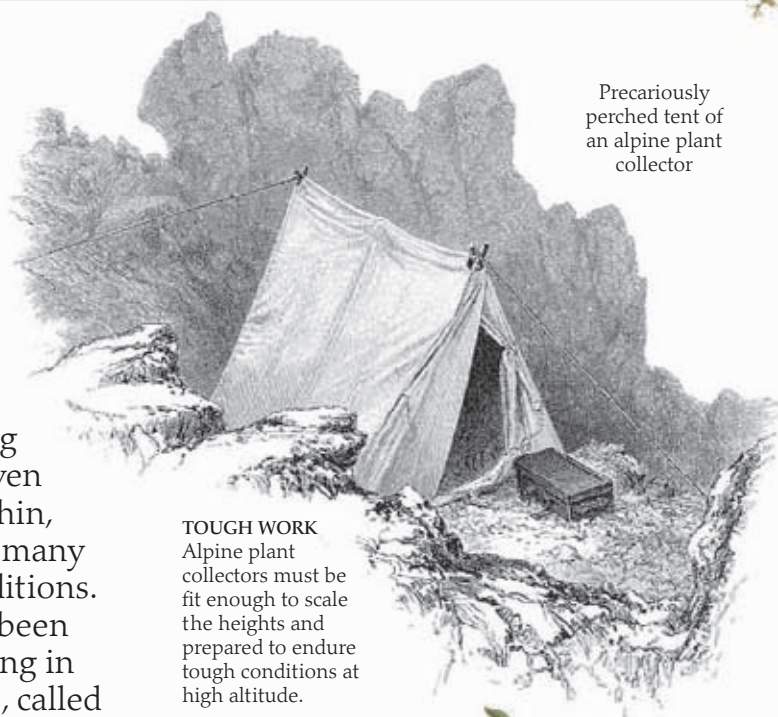
PLANT CUSHIONS

This dwarf hebe from New Zealand is an evergreen plant with small tough leaves that can withstand sharp frost. It grows in dense cushions that trap heat, prevent wind damage, and reduce water loss. These cushions are covered in white flowers every spring.



Dwarf hebe

Precariously perched tent of an alpine plant collector



TOUGH WORK

Alpine plant collectors must be fit enough to scale the heights and prepared to endure tough conditions at high altitude.